Lab 4

2DX3

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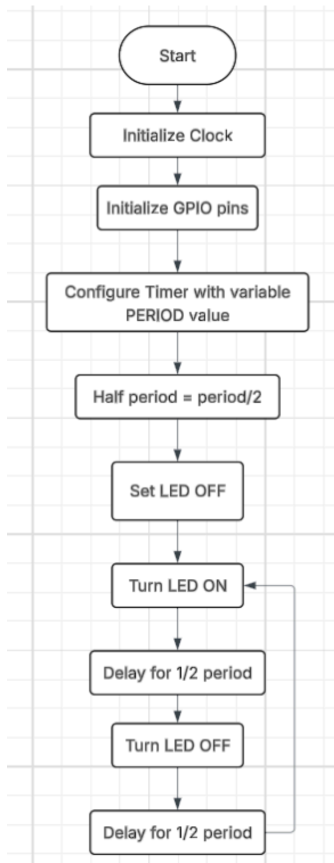
As a future member of the engineering profession, the student is responsible for performing the required work in an honest manner, without plagiarism and cheating. Submitting this work with my name and student number is a statement and understanding that this work is our own and adheres to the Academic Integrity Policy of McMaster University and the Code of Conduct of the Professional Engineers of Ontario.

1. **Draw a flowchart for an ARM microcontroller program to flash an LED with a 50% duty-cycle. Set the period of one cycle as a variable to be later defined.**




2. **Based upon your flowchart, write a C program to flash the onboard LED (provide code in document below). Assuming a 50% duty-cycle, the period of one cycle is defined based upon the least-significant-digit (LSD) of your student number. Include structured and commented source code (must have your name and student number) in your prelab document**

LSD 0,3,6,9: cycle period is 0.50 s
LSD 1,4,7: cycle period is 1.00 s
LSD 2,5,8: cycle period is 2.00 s

*Note: Lab partner's LSD is 3, therefore same cycle period - nothing in code changes*

```
//Name: Claresta
//Student Number: 400582489
//LSD = 9

#include <stdint.h>
```

```c
#include <msp432e401y.h>

#define CYCLE_PERIOD_MS 500 //since LSD is 9
#define DELAY_VALUE (CYCLE_PERIOD_MS / 2)   //since 50 percent of duty
cycle

void GPIO_Init(void);
void Delay_ms(uint32_t ms);

int main(void)
{
    GPIO_Init();

    while(1)    //infinite loop for LED to flash continuously
    {
        GPION-> DATA |= (1 << 1);   //LED ON
        Delay_ms(DELAY_VALUE);      //wait for 250 ms
        GPION-> DATA &= ~(1 << 1);  //LED OFF
        Delay_ms(DELAY_VALUE);      //wait for 250 ms
    }
}

void GPIO_Init(void)
{
    SYSCTL->RCGCGPIO |= (1 << 12);      //enable clock for Port N
    while((SYSCTL->PRGPIO & (1<<12)) == 0);     //wait for Port N to be
ready
    GPION -> DIR |= (1 << 1);       //set PN1 as output
    GPION -> DEN |= (1 << 1);       //enable digital function for PN1
}

void Delay_ms (uint32_t ms)
{
    uint32_t i;
    for(i=0; i<(ms*1600); i++)  //recall PIOSC, how much it runs
    {
        __asm("nop");
    }
}
```

[redacted]

3. **If a stepper motor were to have 36 steps per rotation, how fast would the motor turn with a 10 ms delay per step? Show calculations and answer in RPM. You may assume memory access time etc. are negligible**

We know 36 steps/rotation and 10 ms delay/step

$$T = 36 \times 10 \text{ ms}$$
$$= 360 \text{ ms}$$

$$T = 0.36 \text{ sec}$$

$$RPS = 1/T$$
$$= 1/0.36$$
$$= 2.78 \text{ rev/s}$$

$$RPM = 2.78 \times 60$$
$$= 166.7 \text{ RPM}$$

Therefore motor would turn 166.7 RPM with a 10 ms delay per step

4. **If a stepper motor were to have 200 steps per rotation, calculate the delay per step for the motor to spin at 60 RPM. You may assume memory access time etc. are negligible**

$$\text{Delay per step} = (1/(RPM/60))/200$$
$$= 5\text{ms}$$

Therefore 5 ms delay per step